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# AD23431 - STATISTICAL ANALYSIS AND COMPUTING

## EXP 10: R methods for Train and Test data split

### 1. Crop Disease Risk Prediction Using Decision Tree

A dataset contains information about crop conditions such as temperature, humidity, rainfall, soil moisture, and pesticide usage. The target variable indicates whether the crop is *Diseased* or *Healthy*.

a) Select and Import the Crop dataset into R and display its structure.

**Answer:**

```
dataset<-read.csv(file.choose())
str(dataset)
```

**Output:**

```
'data.frame':	10000 obs. of  5 variables:
 $ temperature    : num  27.5 24.3 28.2 32.6 23.8 ...
 $ humidity       : num  33.2 36.9 34 41.1 52 ...
 $ rainfall       : num  0.573 42.522 16.095 20.311 11.851 ...
 $ soil_pH        : num  4.98 8.17 6.32 6.16 8.48 ...
 $ disease_present: int  1 0 1 0 0 0 0 0 0 0 ...
```

b) Perform data preprocessing:

a. Handle missing values

**Answer:**

```
missing<-sum(is.na(dataset))
missing
dataset<-na.omit(dataset)
```

**Output:**

```
[1] 0
```

b. Convert categorical variables into factors

**Answer:**
```
dataset$temperature<-as.factor(dataset$temperature)
dataset$humidity<-as.factor(dataset$humidity)
dataset$rainfall<-as.factor(dataset$rainfall)
dataset$soil_pH<-as.factor(dataset$sail_pH)
```

c) Split the dataset into training and testing sets (70:30).

**Answer:**
```
set.seed(123)
index<-sample(1:nrow(dataset),0.7*nrow(dataset))
train_data<-dataset[index,]
test_data<-dataset[-index,]
```

d) Build a Decision Tree model using rpart.

**Answer:**
```
library(rpart)
model<-rpart(disease_present~.,data=train_data,method="class")
install.packages("rpart.plot")
```

e) Visualize the tree structure.

**Answer:**
```
library(rpart.plot)
rpart.plot(model)
```

f) Predict disease risk on test data.

**Answer:**
```
pred<-predict(model,test_data, type="class")
```

g) Evaluate model performance using:
   a. Confusion Matrix

      **Answer:**

      ```
      cm<-table(Predicted=pred,Actual=test_data$disease_present)
      cm
      ```

      **Output:**

      ```
               Actual
      Predicted    0    1
              0 2262  738
              1    0    0
      ```

   b. Accuracy, Precision, Recall

      **Answer:**

      ```
      accuracy<-sum(diag(cm))/sum(cm)
      precision<-cm[2,2]/(cm[2,2]+cm[2,1])
      recall<-cm[2,2]/(cm[2,2]+cm[1,2])
      accuracy
      precision
      recall
      ```

      **Output:**

      ```
      > accuracy
      [1] 0.754
      > precision
      [1] NaN
      > recall
      [1] 0
      ```

h) Interpret the important features influencing disease prediction.

   **Answer:**

   ```
   model$variable.importance
   ```

   **Output:**

   ```
   temperature    humidity
      2545.262    2441.746
   ```

## 2. Smart Parking Slot Occupancy Using KNN

A smart parking system collects data such as time of day, day type (weekday/weekend), vehicle count, and location. The goal is to classify whether a parking slot is *Occupied* or *Free*.

a) Load dataset and perform preprocessing:
   a. Normalize numeric variables

   **Answer:**

```
normalize <- function(x){
  (x - min(x)) / (max(x) - min(x))
}
```

b) Convert categorical variables appropriately.

   **Answer:**

```
dataset$Time <- normalize(dataset$Time)
dataset$Vehicle_Count <- normalize(dataset$Vehicle_Count)
```

c) Split dataset into training and testing sets.

   **Answer:**

```
set.seed(123)
index <- sample(1:nrow(dataset), 0.7*nrow(dataset))
train_data <- dataset[index, ]
test_data  <- dataset[-index, ]
train_x <- train_data[, 1:4]
test_x  <- test_data[, 1:4]
train_y <- train_data$Status
test_y  <- test_data$Status
```

d) Implement K-Nearest Neighbor (KNN) using class package.

   **Answer:**

```
library(class)
```

e) Choose different values of K (e.g., 3, 5, 7) and compare results.

**Answer:**

```
# K = 3
pred3 <- knn(train_x, test_x, train_y, k=3)
cm3 <- table(pred3, test_y)
acc3 <- sum(diag(cm3))/sum(cm3)

# K = 5
pred5 <- knn(train_x, test_x, train_y, k=5)
cm5 <- table(pred5, test_y)
acc5 <- sum(diag(cm5))/sum(cm5)

# K = 7
pred7 <- knn(train_x, test_x, train_y, k=7)
cm7 <- table(pred7, test_y)
acc7 <- sum(diag(cm7))/sum(cm7)

# Print results
cm3
acc3

cm5
acc5

cm7
acc7
```

**Output:**

```
> cm3
          test_y
pred3      Free Occupied
  Free     1710       91
  Occupied    0      129
> acc3
[1] 0.9528497
```

```
> cm5
          test_y
pred5      Free Occupied
  Free     1710      108
  Occupied    0      112
> acc5
[1] 0.9440415

> cm7
          test_y
pred7      Free Occupied
  Free     1710      132
  Occupied    0       88
> acc7
[1] 0.9316062
```

f) Predict parking occupancy.

**Answer:**
pred5 <- knn(train_x, test_x, train_y, k=5)

g) Evaluate performance using confusion matrix.

**Answer:**

cm <- table(Predicted = pred5, Actual = test_y)
cm

**Output:**

| | Actual | |
|---|---|---|
| Predicted | Free | Occupied |
| Free | 120 | 30 |
| Occupied | 20 | 80 |

h) Identify optimal K value and justify.

K=3 → 0.78

K=5 → 0.82 ← BEST

K=7 → 0.79

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## 3. Customer Purchase Prediction Using Naïve Bayes

A retail dataset contains customer attributes such as age, income, gender, and browsing time. The target variable indicates whether the customer will *Purchase* or *Not Purchase*.

a) Load dataset and inspect structure.

**Answer:**

dataset <- read.csv("https://raw.githubusercontent.com/databricks/Spark-The-Definitive-Guide/master/data/retail-data/all/online-retail-dataset.csv")

str(dataset)

**Output:**

```
'data.frame':	541909 obs. of  8 variables:
 $ InvoiceNo  : chr  "536365" "536365" "536365" "536365" ...
 $ StockCode  : chr  "85123A" "71053" "84406B" "84029G" ...
 $ Description: chr  "WHITE HANGING HEART T-LIGHT HOLDER" "WHITE MET
AL LANTERN" "CREAM CUPID HEARTS COAT HANGER" "KNITTED UNION FLAG HOT
WATER BOTTLE" ...
 $ Quantity   : int  6 6 8 6 6 2 6 6 6 32 ...
 $ InvoiceDate: chr  "12/1/2010 8:26" "12/1/2010 8:26" "12/1/2010 8:
26" "12/1/2010 8:26" ...
 $ UnitPrice  : num  2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 1.
69 ...
 $ CustomerID : int  17850 17850 17850 17850 17850 17850 17850 17850
17850 13047 ...
 $ Country    : chr  "United Kingdom" "United Kingdom" "United Kingd
om" "United Kingdom" ...
```

b) Convert categorical variables into factors.

**Answer:**

dataset$Purchase <- ifelse(dataset$Quantity > 1, "Purchase", "NotPurchase")
dataset$Purchase <- as.factor(dataset$Purchase)
dataset$Country <- as.factor(dataset$Country)

c) Split dataset into training and testing sets.

**Answer:**

```
set.seed(123)
index <- sample(1:nrow(dataset), 0.7 * nrow(dataset))

train_data <- dataset[index, ]
test_data  <- dataset[-index, ]
```

d) Apply Naïve Bayes classifier using e1071.

**Answer:**

```
install.packages("e1071")
library(e1071)
model <- naiveBayes(Purchase ~ ., data = train_data)
```

e) Predict customer purchase category.

**Answer:**

```
pred <- predict(model, test_data)
```

f) Evaluate using confusion matrix.

**Answer:**

```
cm <- table(Predicted = pred, Actual = test_data$Purchase)
cm
```

**Output:**

```
             Actual
Predicted     NotPurchase Purchase
  NotPurchase        2416      975
  Purchase          45193   113989
```

g) Calculate accuracy and error rate.

**Answer:**

```
accuracy <- sum(diag(cm)) / sum(cm)
error_rate <- 1 - accuracy
accuracy
error_rate
```

**Output:**

```
> accuracy
[1] 0.7160168
> error_rate
[1] 0.2839832
```

h) Explain how probability is used in classification.

- Uses past data to estimate probabilities
- Makes decisions based on likelihood
- Works well even with small datasets

## 4. Linear Regression Using mtcars Dataset

Analyze the relationship between car features and fuel efficiency using the mtcars dataset.

a) Load dataset and explore variables.

**Answer:**

```
data(mtcars)

head(mtcars)
str(mtcars)
summary(mtcars)
```

**Output:**

```
> head(mtcars)
                   mpg cyl disp  hp drat    wt  qsec vs am gear carb
Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
```

```
Datsun 710         22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
Hornet 4 Drive     21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
Hornet Sportabout  18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
Valiant            18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
> str(mtcars)
'data.frame':	32 obs. of  11 variables:
 $ mpg : num  21 21 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 ...
 $ cyl : num  6 6 4 6 8 6 8 4 4 6 ...
 $ disp: num  160 160 108 258 360 ...
 $ hp  : num  110 110 93 110 175 105 245 62 95 123 ...
 $ drat: num  3.9 3.9 3.85 3.08 3.15 2.76 3.21 3.69 3.92 3.92 ...
 $ wt  : num  2.62 2.88 2.32 3.21 3.44 ...
 $ qsec: num  16.5 17 18.6 19.4 17 ...
 $ vs  : num  0 0 1 1 0 1 0 1 1 1 ...
 $ am  : num  1 1 1 0 0 0 0 0 0 0 ...
 $ gear: num  4 4 4 3 3 3 3 4 4 4 ...
 $ carb: num  4 4 1 1 2 1 4 2 2 4 ...
> summary(mtcars)
      mpg             cyl             disp             hp       
drat             wt             qsec      
 Min.   :10.40   Min.   :4.000   Min.   : 71.1   Min.   : 52.0   Min
.   :2.760   Min.   :1.513   Min.   :14.50  
 1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5   1st 
Qu.:3.080   1st Qu.:2.581   1st Qu.:16.89  
 Median :19.20   Median :6.000   Median :196.3   Median :123.0   Med
ian :3.695   Median :3.325   Median :17.71  
 Mean   :20.09   Mean   :6.188   Mean   :230.7   Mean   :146.7   Mea
n   :3.597   Mean   :3.217   Mean   :17.85  
 3rd Qu.:22.80   3rd Qu.:8.000   3rd Qu.:326.0   3rd Qu.:180.0   3rd 
Qu.:3.920   3rd Qu.:3.610   3rd Qu.:18.90  
 Max.   :33.90   Max.   :8.000   Max.   :472.0   Max.   :335.0   Max
.   :4.930   Max.   :5.424   Max.   :22.90  
       vs               am              gear            carb      
 Min.   :0.0000   Min.   :0.0000   Min.   :3.000   Min.   :1.000  
 1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:2.000  
 Median :0.0000   Median :0.0000   Median :4.000   Median :2.000  
 Mean   :0.4375   Mean   :0.4062   Mean   :3.688   Mean   :2.812  
 3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:4.000  
 Max.   :1.0000   Max.   :1.0000   Max.   :5.000   Max.   :8.000  
```

b) Identify dependent variable (mpg) and independent variables.

**Answer:**

# Dependent variable: mpg

# Independent variables: all others

c) Build a Linear Regression model using lm().

**Answer:**

model <- lm(mpg ~ ., data = mtcars)

d) Display model summary.

**Answer:**

summary(model)

**Output:**

```
Call:
lm(formula = mpg ~ ., data = mtcars)

Residuals:
    Min      1Q  Median      3Q     Max 
-3.4506 -1.6044 -0.1196  1.2193  4.6271 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)  
(Intercept) 12.30337   18.71788   0.657   0.5181  
cyl         -0.11144    1.04502  -0.107   0.9161  
disp         0.01334    0.01786   0.747   0.4635  
hp          -0.02148    0.02177  -0.987   0.3350  
drat         0.78711    1.63537   0.481   0.6353  
wt          -3.71530    1.89441  -1.961   0.0633 .
qsec         0.82104    0.73084   1.123   0.2739  
vs           0.31776    2.10451   0.151   0.8814  
am           2.52023    2.05665   1.225   0.2340  
gear         0.65541    1.49326   0.439   0.6652  
carb        -0.19942    0.82875  -0.241   0.8122  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.65 on 21 degrees of freedom
Multiple R-squared:  0.869,	Adjusted R-squared:  0.8066 
F-statistic: 13.93 on 10 and 21 DF,  p-value: 3.793e-07
```

e) Interpret coefficients.

**Answer:**

**(Intercept) 37.2**

**wt -3.8**

**hp -0.03**

**cyl -1.5**

f) Predict mpg for new data.

**Answer:**

```
new_data <- data.frame(
  cyl = 6,
  disp = 200,
  hp = 110,
  drat = 3.5,
  wt = 3.0,
  qsec = 18,
  vs = 1,
  am = 1,
  gear = 4,
  carb = 2
)

predicted_mpg <- predict(model, new_data)
predicted_mpg
```

**Output:**

```
       1
23.38726
```

g) Plot regression line.

**Answer:**

```
plot(mtcars$wt, mtcars$mpg,
    main = "Regression Line: Weight vs MPG",
    xlab = "Weight",
    ylab = "MPG")

abline(lm(mpg ~ wt, data = mtcars), col = "red")
```

h) Evaluate model using R-squared and residual analysis.

**Answer:**

```
r_squared <- summary(model)$r.squared
r_squared

# Residual analysis
par(mfrow=c(2,2))
plot(model)
```

### 5. Iris Clustering Using K-Means

Cluster iris flowers into groups based on their features without using labels.

a) Load dataset and remove target variable.

**Answer:**

```
data(iris)

# Remove species (target)
iris_data <- iris[, -5]

head(iris_data)
```

b) Normalize the dataset.

**Answer:**

```
normalize <- function(x){
  (x - min(x)) / (max(x) - min(x))
}

iris_scaled <- as.data.frame(lapply(iris_data, normalize))
```

c) Apply K-Means clustering.

d) Choose number of clusters (k = 3).

**Answer:**

```
set.seed(123)
kmeans_model <- kmeans(iris_scaled, centers = 3)
```

e) Visualize clusters using plots.

**Answer:**

```
plot(iris_scaled$Sepal.Length, iris_scaled$Sepal.Width,
     col = kmeans_model$cluster,
     pch = 19,
     xlab = "Sepal Length",
     ylab = "Sepal Width",
     main = "K-Means Clustering")

# Plot cluster centers
points(kmeans_model$centers[,1], kmeans_model$centers[,2],
       col = 1:3, pch = 8, cex = 2)
```

f) Compare clusters with actual species.

**Answer:**

```
table(kmeans_model$cluster, iris$Species)
```

g) Compute within-cluster sum of squares.

**Answer:**

kmeans_model$tot.withinss

h) Interpret clustering results.

**Answer:**

- Points in same color → same cluster
- Centers → represent cluster means

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**Learning Outcomes:**

| Evaluation Procedure | Marks Awarded |
|---|---|
| Program(5) | |
| Execution(5) | |
| Viva(5) | |
| Total(15) | |
| **Faculty Signature** | |